

Spatial Representation and CRS

Coordinate reference systems, datums,
projections, vector, raster

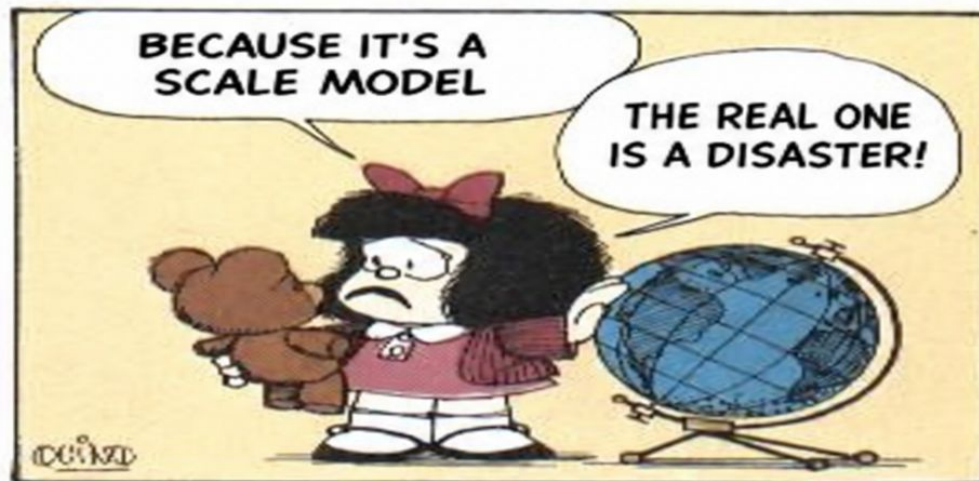
Édipo Cremon
Rennan Marujo



What we will build



- Represent geographic phenomena as objects or fields.
- Connect coordinates to datums, projections, units, and metadata.
- Select a CRS for measurement and data integration.
- Detect common CRS mistakes before analysis.



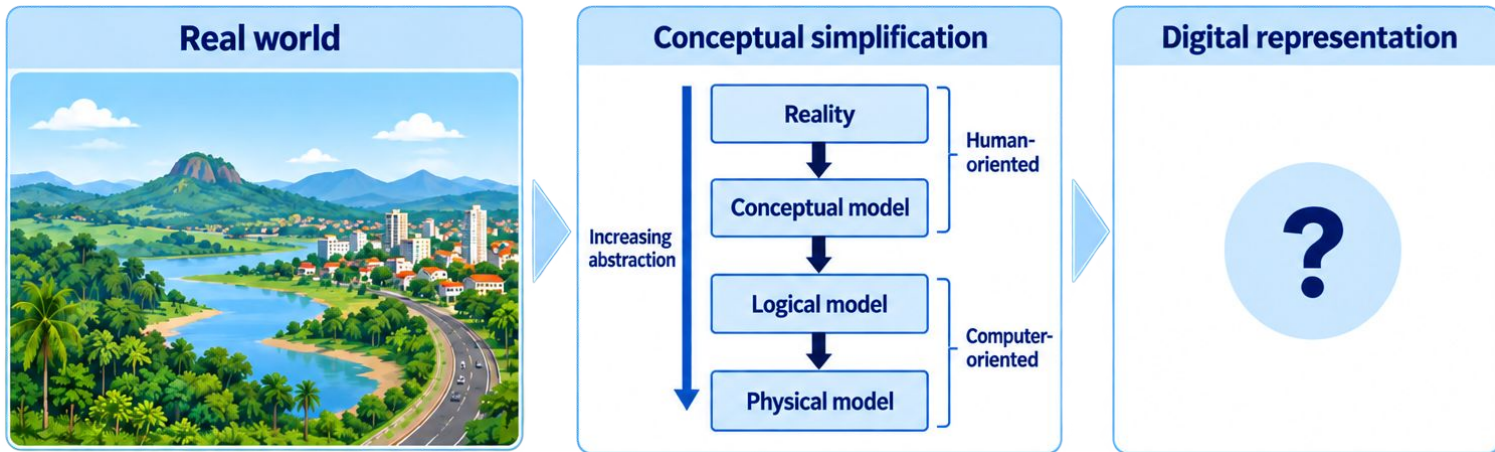
Editado - <http://naahdolls.blogspot.com>

From real world to digital

The computer does not store the world itself.

It stores a **simplified representation** of objects, locations, and attributes.

- The real world is continuous, complex, and detailed.
- A computer needs structured representations.
- Geographic data begins with **selection, abstraction, and encoding**.



A map or dataset is a model of reality, not reality itself.

Geospatial data



- **Raster Representation**
- **Vector Representation**

Raster

Represents the real world using a regular grid of **cells or pixels**.

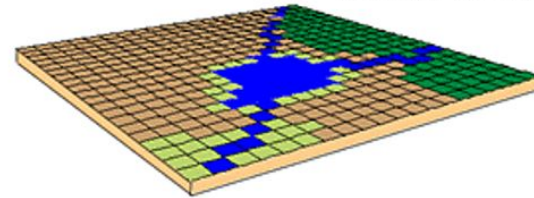
Each cell is defined by its position in the **rows and columns** of the grid.

Vector

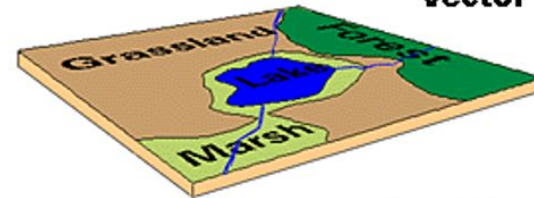
Represents real-world features using **coordinates**.

The basic spatial unit is defined by **x, y coordinate pairs**.

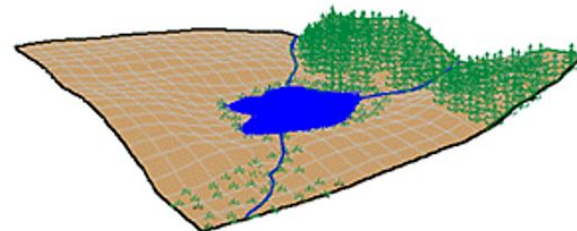
Raster / Image



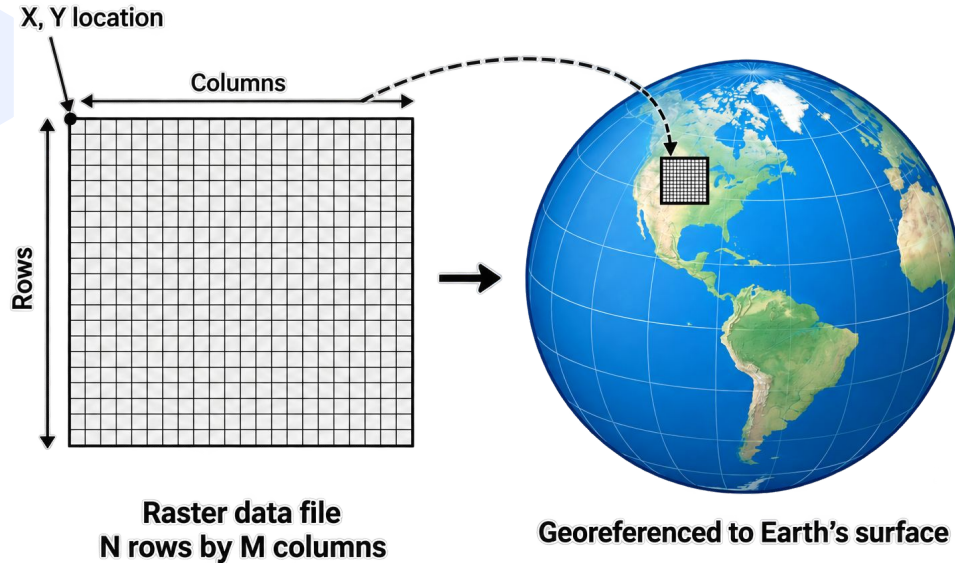
Vector



Real World



Raster: Cells, Extent, Resolution



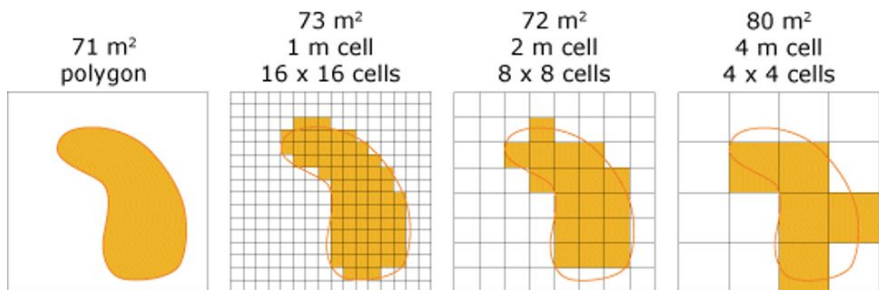
A raster grid is defined by:

- number of rows and columns
- spatial extent and origin

Changing the CRS* of a raster usually requires resampling.

*CRS = Coordinate Reference System

Raster: Cells, Extent, Resolution



- Smaller cell size
- Higher resolution
- Higher feature spatial accuracy
- Slower display
- Slower processing
- Larger file size

- Larger cell size
- Lower resolution
- Lower feature spatial accuracy
- Faster display
- Faster processing
- Smaller file size

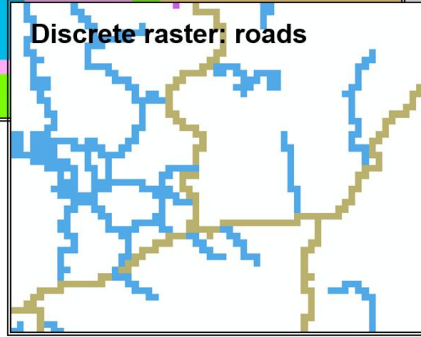
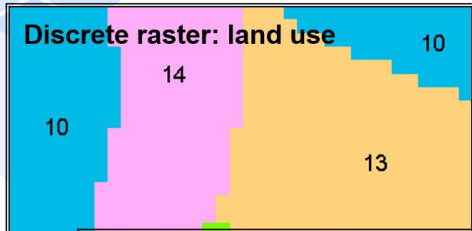
A raster grid is defined by:

- number of rows and columns
- spatial extent and origin
- cell size on the ground

Changing the CRS* of a raster usually requires resampling.

*CRS = Coordinate Reference System

Raster: Cells, Extent, Resolution



A raster grid is defined by:

- number of rows and columns
- spatial extent and origin
- cell size on the ground
- value type: continuous or categorical

Changing the CRS* of a raster usually requires resampling.

*CRS = Coordinate Reference System

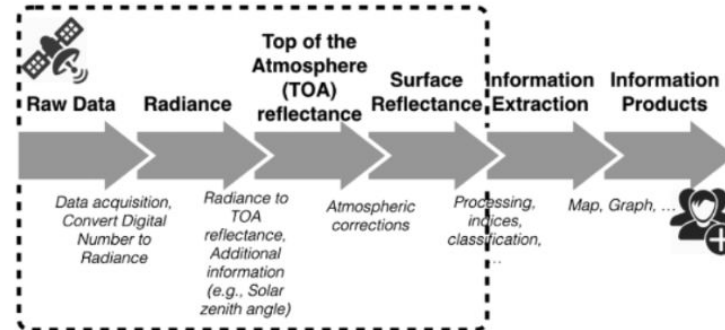
Data Cubes - BDC exemple

ARD and EO Data Cubes for the Entire Brazilian Territory

2 Petabytes (PB)
of Remote Sensing
Imagery

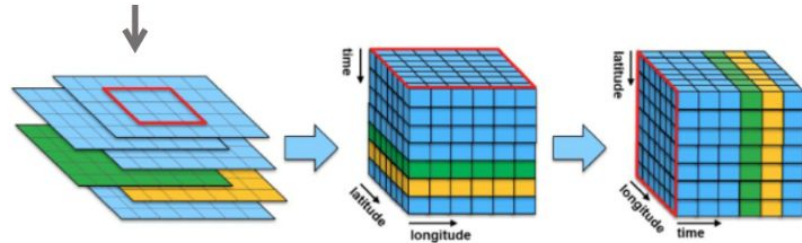


BDC Hierarchical Tiling System



Analysis Ready Data production

Source: [Giuliani et al, 2017]



Collections of ARD
satellite imagens

Multidimensional Data cubes

Source: [Kopp et al, 2019]



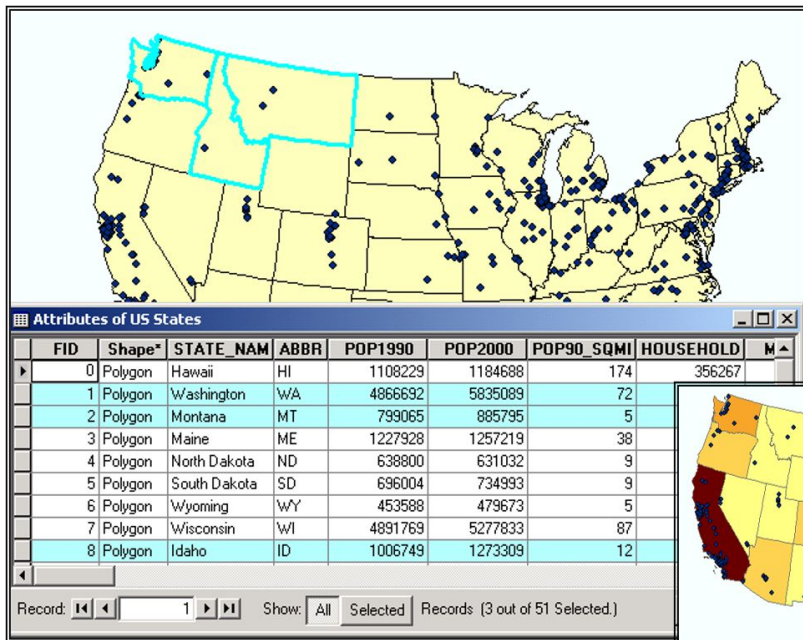
Common Raster Data Formats



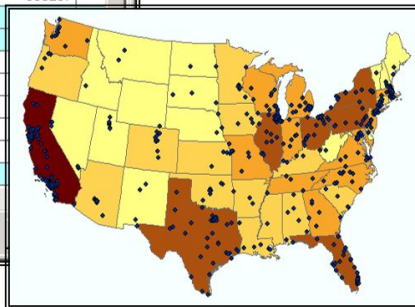
Format	Common use	Advantages	Limitations	Introduced
GeoTIFF / COG*	GIS, Earth observation, DEMs	Georeferencing embedded; broad software support; compression; COG enables efficient remote access	Storage and I/O depend on compression and tiling configuration; limited multidimensional support; COG requires specific internal organization	1995 / 2023*
NetCDF	Climate, oceanography, time series, data cubes	Multidimensional arrays; rich metadata; well suited to space-time data	More complex structure; performance depends on chunking and organization	1989
HDF5	Satellite products, scientific multidimensional data	Hierarchical structure; compression; large multidimensional datasets	Complex structure; georeferencing conventions vary among products	1998
JPEG 2000	Satellite imagery, compressed imagery	Lossless or lossy compression; multiresolution access; high compression efficiency	Slower encoding/decoding; less universal GIS support than GeoTIFF	2001
Zarr	Large data cubes, distributed and remote array processing	Chunked N-D arrays; parallel access; efficient subset reading	Many-file/object structure; geospatial conventions still evolving	2016

*COG = Cloud Optimized GeoTIFF

Geospatial data: vector

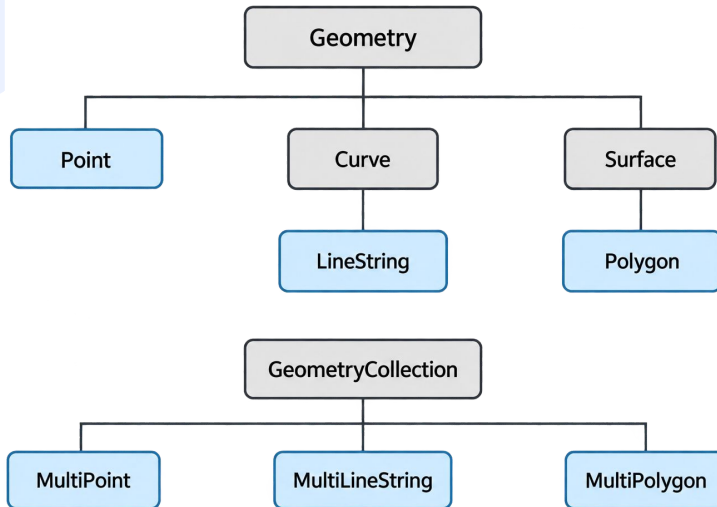


Each vector feature is linked to an attribute table that store information about that feature.

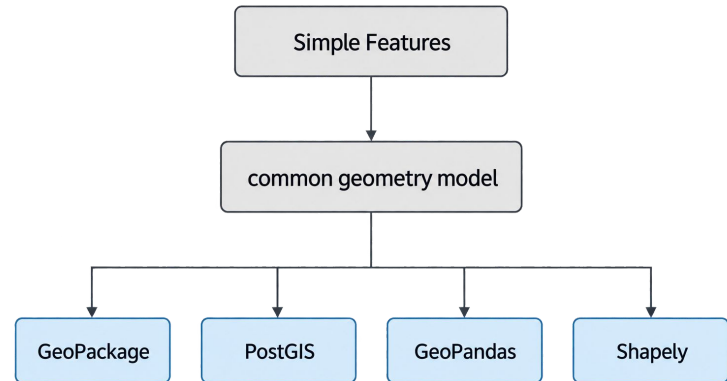


Simple Features: A Common Vector Model

ISO 19125 / OGC Simple Feature Access



- Defines a common model for representing **vector geometries**.
- Core geometry types include **Point, Curve, Surface, and GeometryCollection**.
- Geometries are associated with a **Coordinate Reference System (CRS)**.
- Provides a common foundation for geospatial software and spatial databases.



Part 1: Common geometry architecture

Part 2: SQL support for storing and querying features

Vector Geometry Types



Single geometries

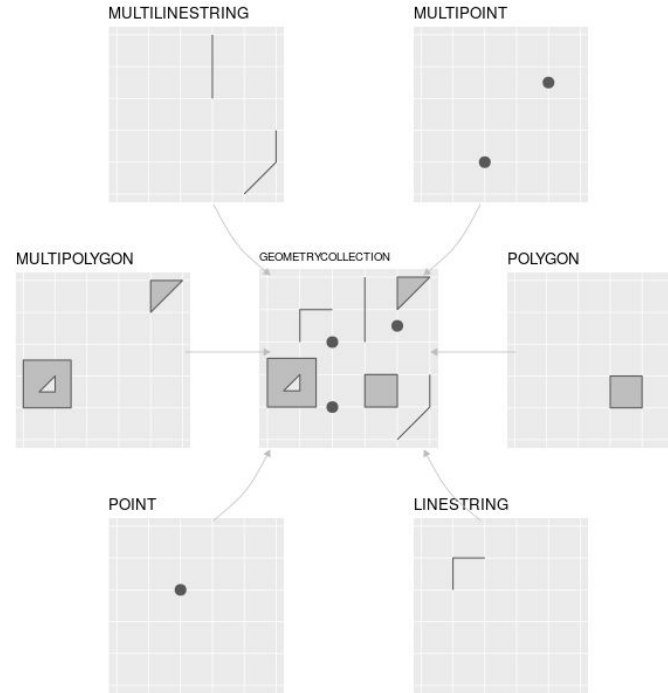
Point · LineString · Polygon

Multi-geometries

MultiPoint · MultiLineString · MultiPolygon

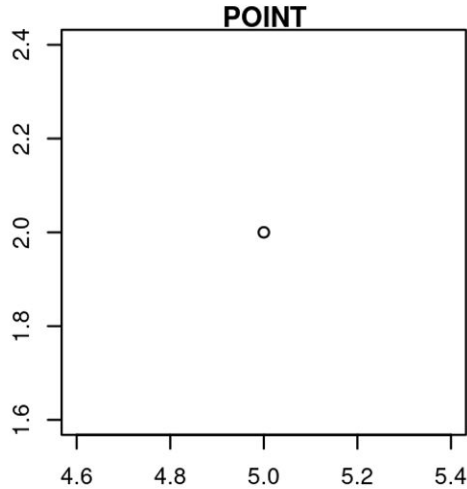
Mixed geometries

GeometryCollection

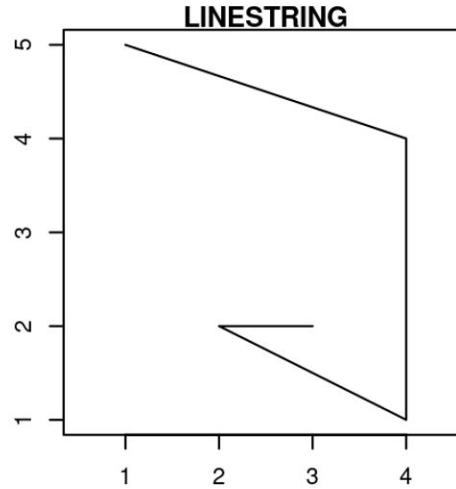


Vector data represent discrete geographic features using geometries defined by coordinates.

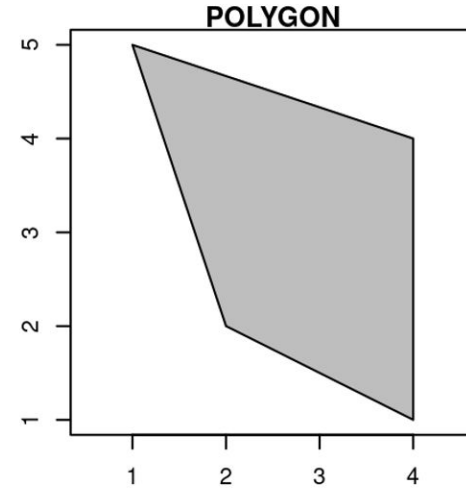
Vector Geometry Types



POINT (5 2)



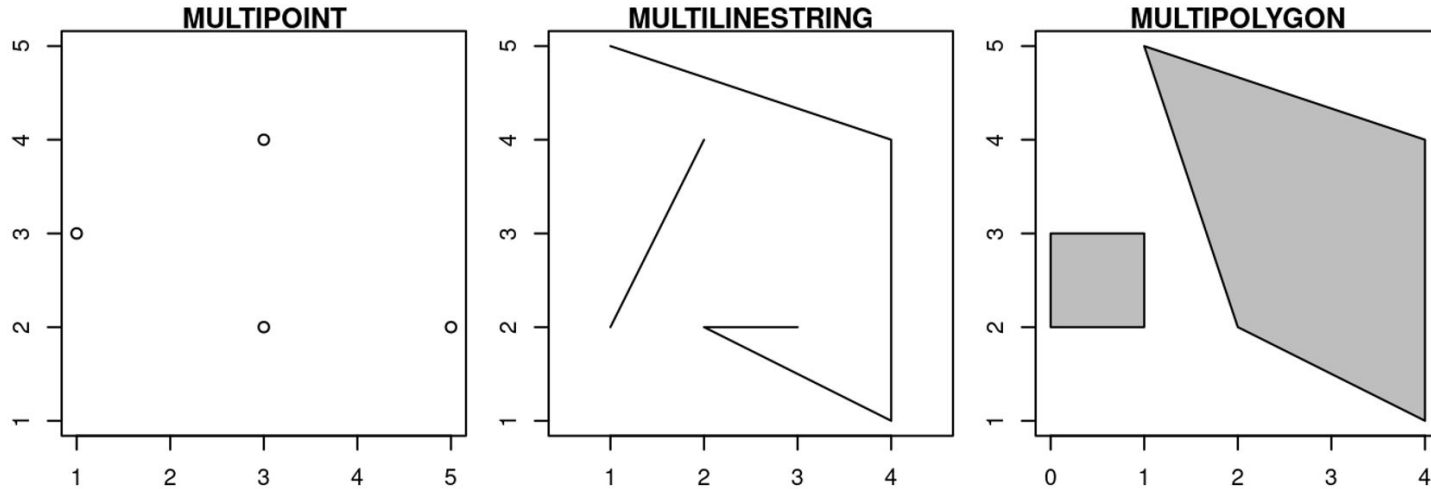
LINESTRING (1 5, 4 4, 4 1, 2 2, 3 2)



POLYGON ((1 5, 2 2, 4 1, 4 4, 1 5))

Vector data represent discrete geographic features using geometries defined by coordinates.

Vector Geometry Types



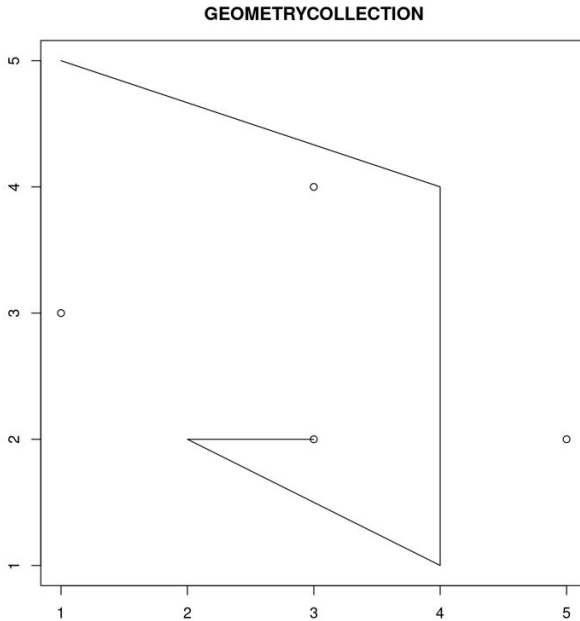
Multipoint: **MULTIPOINT (5 2, 1 3, 3 4, 3 2)**

Multilinestring: **MULTILINESTRING ((1 5, 4 4, 4 1, 2 2, 3 2), (1 2, 2 4))**

Multipolygon: **MULTIPOLYGON (((1 5, 2 2, 4 1, 4 4, 1 5), (0 2, 1 2, 1 3, 0 3, 0 2)))**

Vector data represent discrete geographic features using geometries defined by coordinates.

Vector Geometry Types



Geometry collection: **GEOMETRYCOLLECTION**
(MULTIPOINT (5 2, 1 3, 3 4, 3 2), LINESTRING (1 5,
4 4, 4 1, 2 2, 3 2))

Vector data represent discrete geographic features using geometries defined by coordinates.

Vector data: beyond geometry

Vector features can be analyzed through their **spatial relationships**.

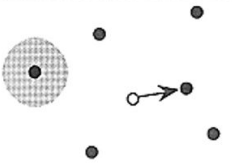
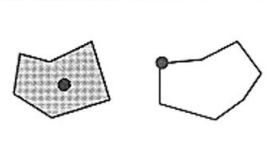
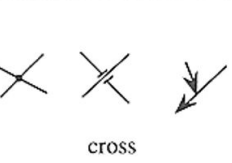
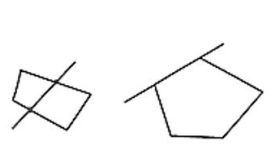
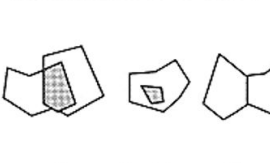
Point - Point	Point - Line	Point - Area
 is within nearest to	 on line nearest to	 in area on area
Line - Line	Line - Area	Area - Area
 intersect cross flow into	 intersect border	 overlap inside adjacent to

Figure 2.4 Topological Relationships Between Spatial Objects

Vector data: beyond geometry

Vector features can be analyzed through their **spatial relationships**.



Format	Best suited for	Strengths	Limitations
GeoPackage	General-purpose GIS	Single file · multiple layers · spatial index · open standard	More complex internal structure
GeoJSON	Web & APIs	Human-readable · simple · web-native	Large files · WGS 84-oriented
Shapefile	Legacy data exchange	Very broad software support	Multiple files · size and attribute limitations
GeoParquet	Geo Data Science	Columnar · compressed · fast analytical I/O	Newer ecosystem

Other formats: FlatGeobuf · KML · File Geodatabase



QUIZ

Can the same geographic phenomenon be represented in different ways? ?

How would you represent a river in a geospatial dataset?

- A. As a line
- B. As a polygon
- C. As raster cells





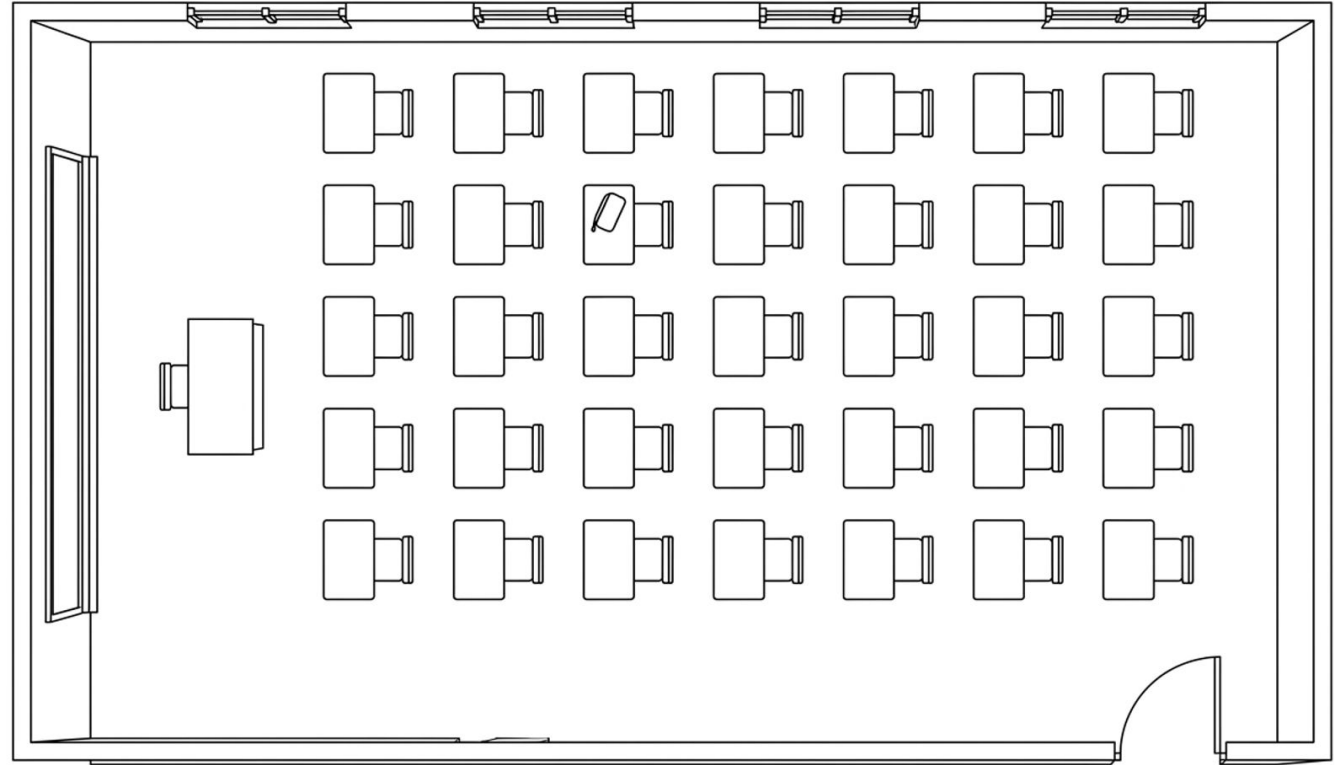
Coordinate Reference System

Locate the pencil case



**A student left a pencil case in a classroom.
How can another student find it?**

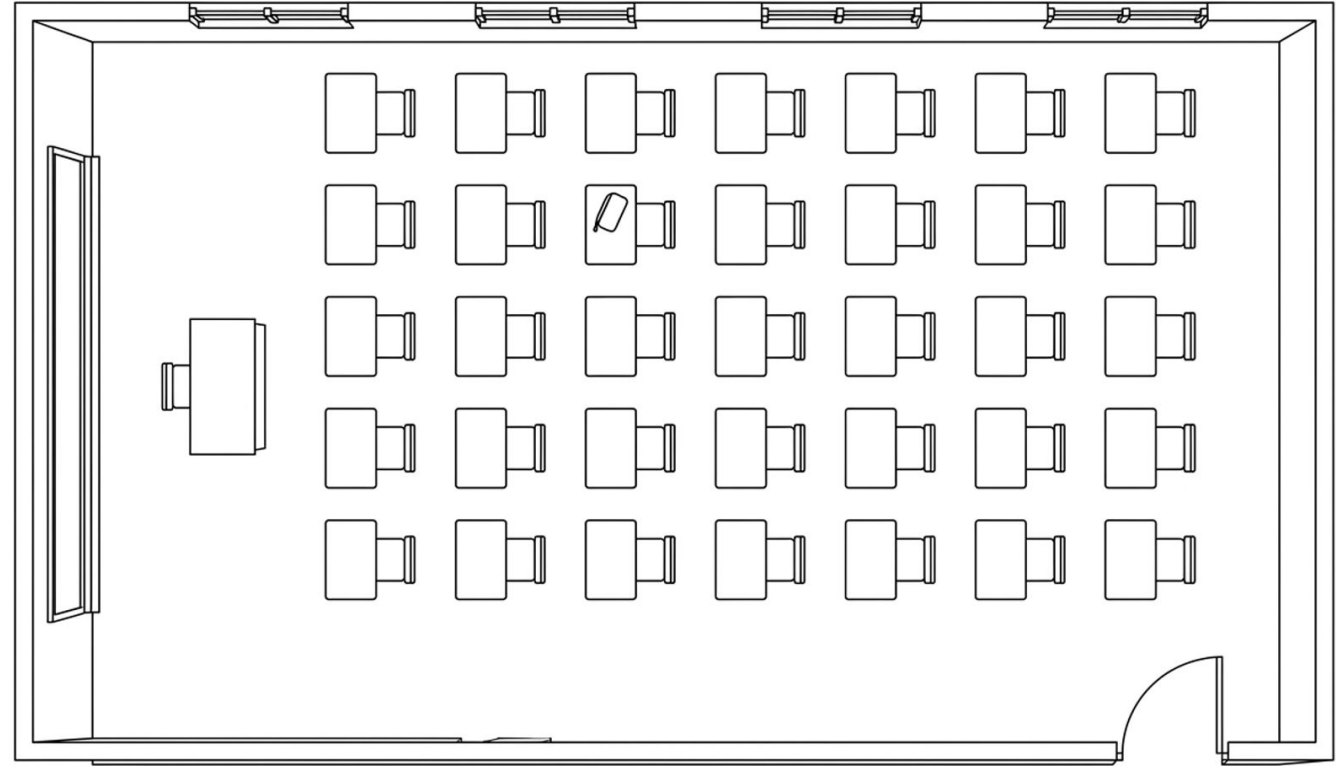
- A shared origin: front-left corner.
- Axes: rows and columns.
- Units: desks, steps, or meters.
- A location becomes an address.



Locate the pencil case

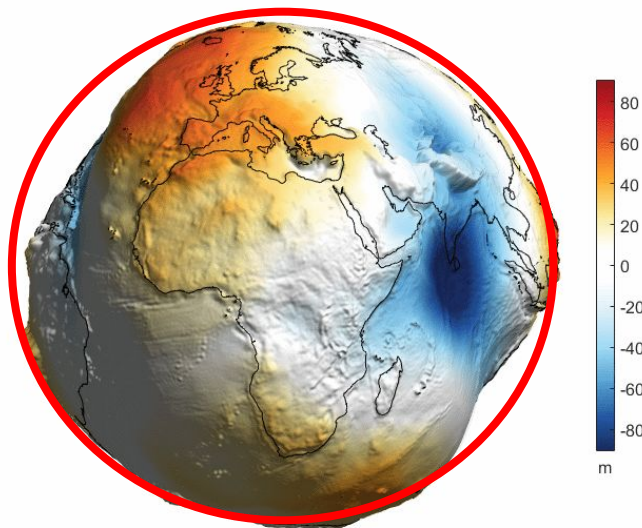


**A student left a pencil case in a classroom.
How can another student find it?**



Geoid, Ellipsoid, Datum

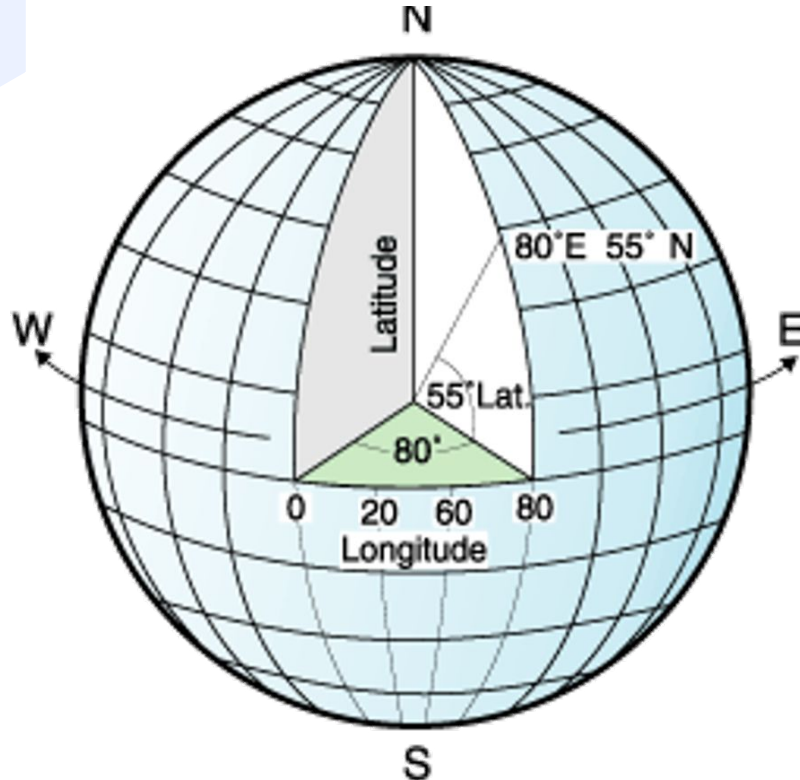
- The geoid follows Earth gravity and mean sea level;
- The ellipsoid provides a smooth calculation surface. The mathematical surface;
- The datum anchors the ellipsoid to Earth.



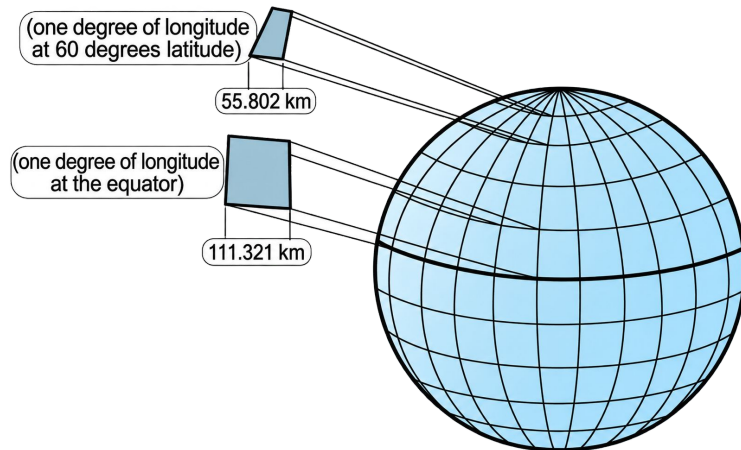
Datum: ellipsoid plus origin, orientation, and realization

Geographic Coordinates

Latitude and longitude are angular coordinates.

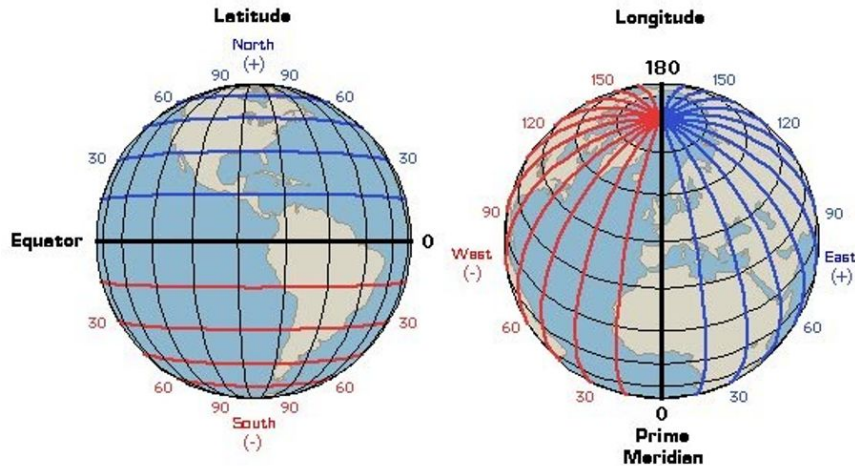


- Latitude: angle north or south of the equator.
- Longitude: angle east or west of Greenwich.
- Common formats: DMS and decimal degrees.
- One degree of longitude changes length with latitude.



Geographic Coordinates

Latitude and longitude are angular coordinates.



- Latitude: angle north or south of the equator.
- Longitude: angle east or west of Greenwich.
- Common formats: DMS and decimal degrees.
- One degree of longitude changes length with latitude.

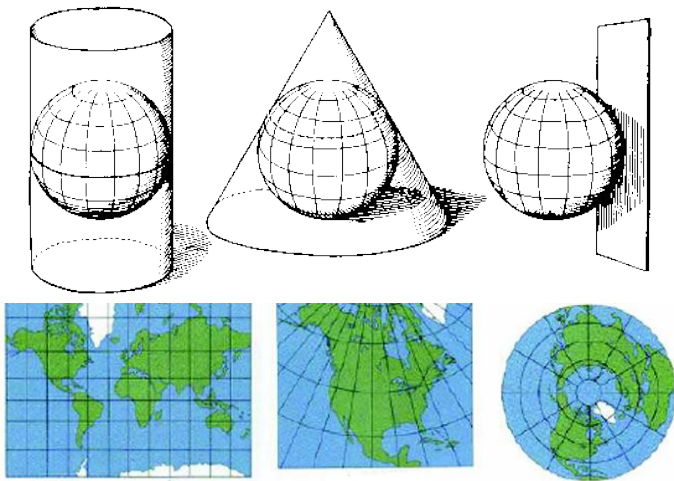
One degree of longitude changes length with latitude.

At the equator: about 111 km

At 60° latitude: about 56 km

Projected Coordinates

Projection transforms latitude and longitude into x and y .



- Projected coordinates use linear units such as meters.
- Each projection preserves some properties better than others.
- Common choices preserve area, angle, distance, or direction for specific contexts.

Geographic CRS



Projection

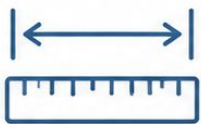


x , y in meters



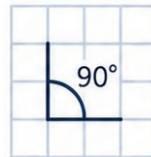
Map projection properties

Different projections preserve different geometric properties



Equidistant

Distances along specific lines are kept at uniform scale



Conformal

Local angles and small shapes are preserved



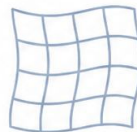
A = 100



A = 100

Equal-area

Area relationships are preserved

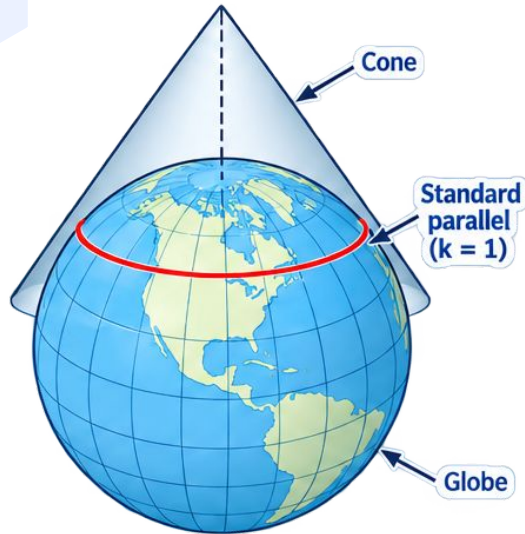


Aphylactic

No single property is fully preserved

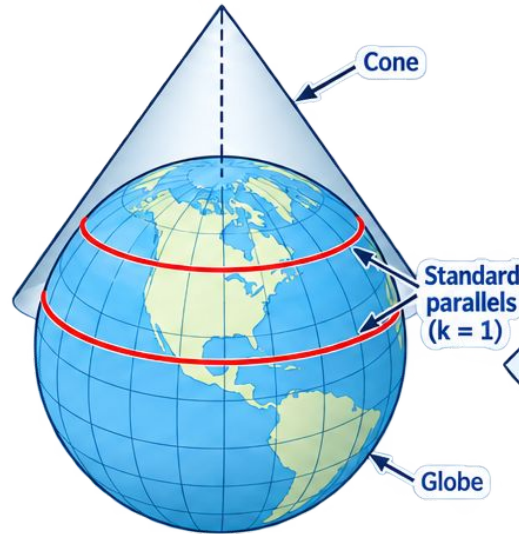
Conic projection

Tangent at one standard parallel



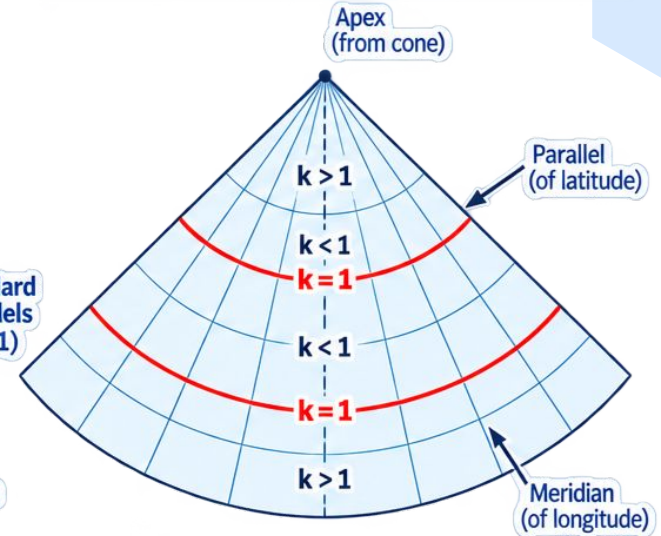
The cone touches the globe at a single standard parallel. Scale is true ($k = 1$) along this parallel.

Secant at two standard parallels



The cone intersects the globe at two standard parallels. Scale is true ($k = 1$) along both parallels, with reduced distortion between them.

Developed conic graticule and scale factor



On a conic projection, the scale factor k varies with latitude. For a secant cone, $k = 1$ at the two standard parallels, $k < 1$ between them, and $k > 1$ outside them. For a tangent cone, $k = 1$ at the single standard parallel, with $k > 1$ elsewhere.

UTM in Practice

Universal Transverse Mercator



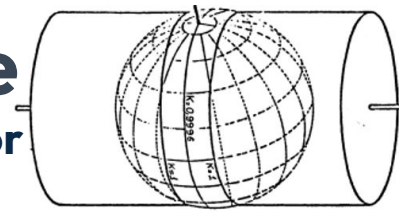
Universal Transverse Mercator

- World divided into 6-degree longitude zones.
- Coordinates reported as easting and northing.
- Good for local and regional work within one zone.
- Brazil spans several UTM zones.

**Example: SIRGAS 2000 / UTM zone 23S uses
EPSG:31983.**

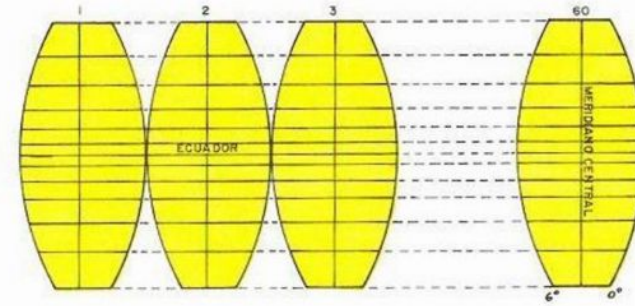
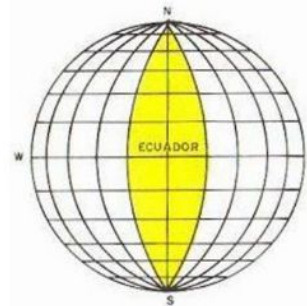
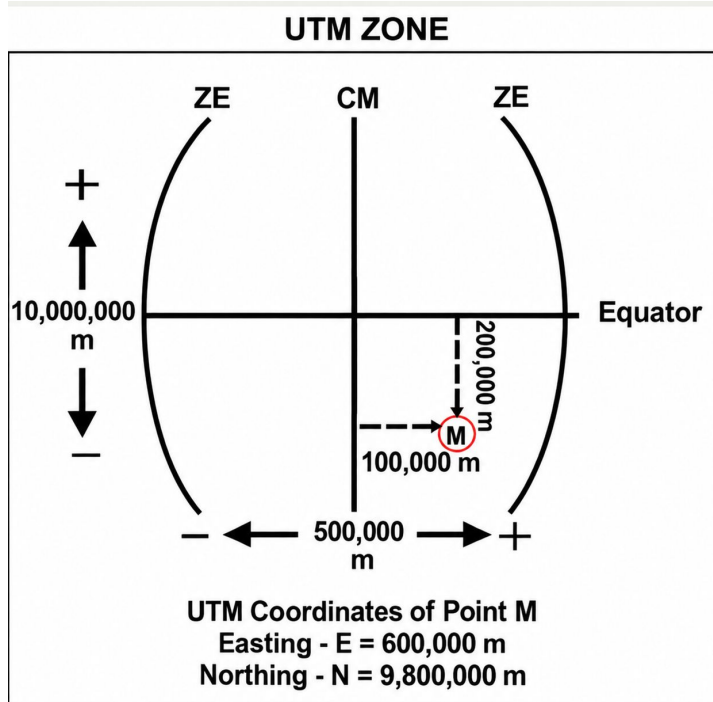
UTM in Practice

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Universal Transverse Mercator

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QUIZ



**A dataset contains the coordinates
 $X = 350000$ and $Y = 7400000$. Can
we determine where the
observations are located?**

- A. Yes, because every coordinate pair defines a unique location on Earth
- B. Yes, if we know whether the data are raster or vector
- C. No, we also need to know the coordinate reference system
- D. No, unless the coordinates are latitude and longitude

EPSG Codes



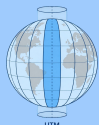
An EPSG code is a unique identifier for a Coordinate Reference System (CRS)

Examples



EPSG:4326
WGS 84

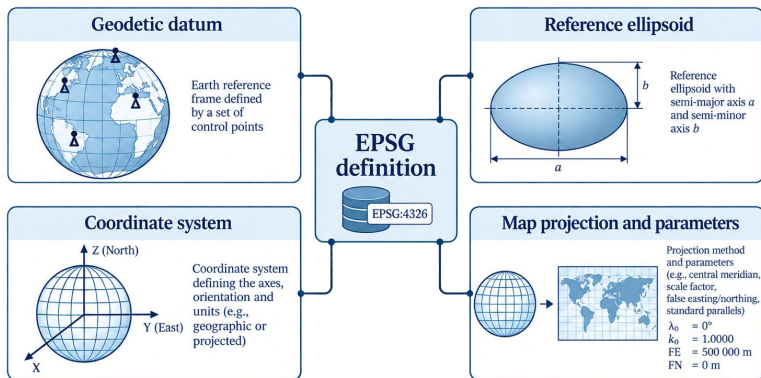
Geographic CRS



EPSG:31983
SIRGAS 2000
UTM zone 23S

Project CRS

What does an EPSG definition include?



- Geodetic datum
- Reference ellipsoid
- Coordinate system
- Map projection and parameters

Used across the geospatial ecosystem



QGIS · ArcGIS



PostGIS



PROJ · GDAL

EPSG is not a map projection. It is an identifier for a complete CRS definition.

Looking up a CRS with EPSG.io

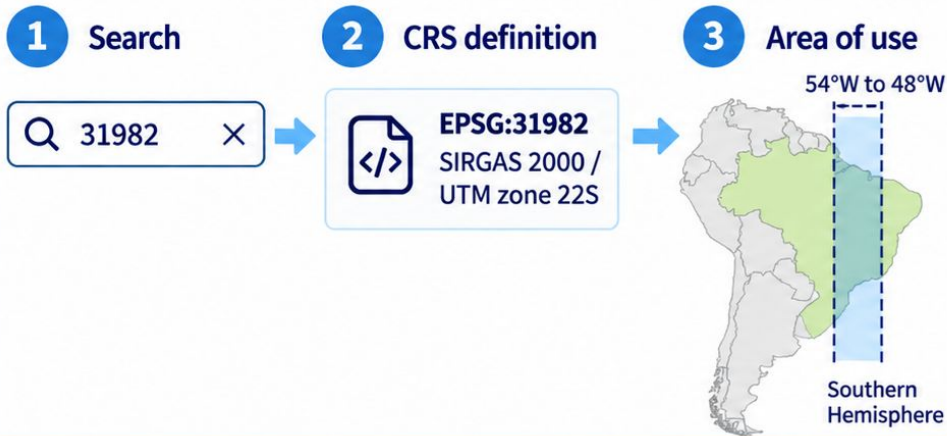
EPSG.io provides a practical interface for searching and inspecting CRS



Search by

-  CRS name
-  EPSG code
-  Region

Example



For each CRS, EPSG.io provides

-  WKT-CRS definition
-  PROJ string
-  Area of use
-  Interactive map
-  Export options

A CRS definition includes where the system is intended to be used.

Common CRS codes



Code	CRS	Type	Typical context
EPSG:4326	WGS 84	Geographic	GNSS, global datasets
EPSG:4674	SIRGAS 2000	Geographic	Brazilian geodetic reference
EPSG:31982	SIRGAS 2000 / UTM 22S	Projected	Regional mapping
EPSG:31983	SIRGAS 2000 / UTM 23S	Projected	Regional mapping
EPSG:31984	SIRGAS 2000 / UTM 24S	Projected	Regional mapping
EPSG:10857	SIRGAS 2000 / Brazil Albers	Projected	National area/statistical analysis/Brazil Data Cube
EPSG:3857	WGS 84 / Pseudo-Mercator	Projected	Web maps



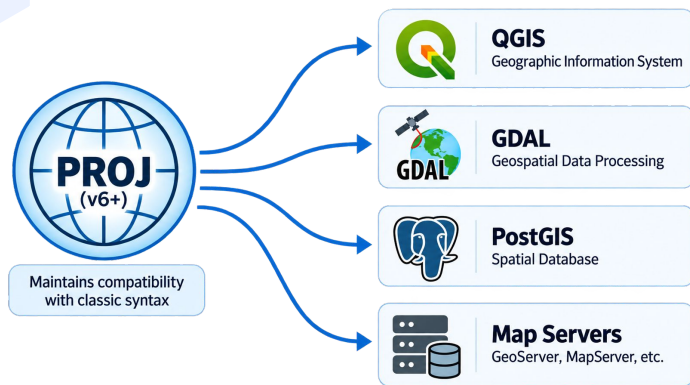
Nigeria example 

**Traditional Nigerian
CRS: Minna / Nigeria Mid
Belt**

There is no universal “best CRS”: location, extent, and purpose guide the choice.

PROJ: Coordinate Transformation Engine

PROJ is an open-source library for coordinate reference system transformations and cartographic projections.



- Originally developed as **PROJ.4**, it later evolved into the modern **PROJ** library.
- Supports transformations between **geographic and projected coordinate systems**.
- Provides the transformation engine used across much of the open-source geospatial ecosystem.



PROJ: Coordinate Transformation Engine

A PROJ string explicitly describes the parameters used to define a CRS or coordinate operation.

```
+proj=utm +zone=23 +south +ellps=GRS80 +units=m +no_defs
```

+proj=utm → Projection method

+zone=23 +south → Zone and hemisphere

+ellps=GRS80 → Reference ellipsoid

+units=m → Coordinate units

EPSG identifies the CRS. PROJ handles its computational definition and transformations.



WKT-2: a standard CRS definition

ISO 19162 · OGC standard



WKT-2 is a structured text format for describing a Coordinate Reference System in detail.

1

Standardized across geospatial software

2

Self-contained CRS description

3

Supports **dynamic datums and epochs**

4

Handles complex coordinate operations

WKT-2 exemple (shortened)

```
PROJCRS["SIRGAS 2000 / UTM  
zone 23S",  
BASEGEOGCRS["SIRGAS 2000",  
DATUM["Sistema de Referencia  
Geocentrico..."],  
ELLIPSOID["GRS 1980", ...]],  
CONVERSION["UTM zone 23S",  
...], CS[Cartesian,2],  
LENGTHUNIT["metre",1]]
```

Verbose

Long hierarchical definitions are difficult to write and read manually.

EPSG identifies the CRS. PROJ enables coordinate operations. WKT-2 provides a detailed, standardized CRS description.

Examples - EPSG 10857 (used in BDC)



WKT-2 exemple

```
PROJCRS["SIRGAS 2000 / Brazil Albers",
  BASEGEOGCRS["SIRGAS 2000",
    DATUM["Sistema de Referencia Geocentrico para las AmericaS 2000",
      ELLIPSOID["GRS 1980",6378137,298.257222101,
        LENGTHUNIT["metre",1]],
      PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
      ID["EPSG",4674]],
    CONVERSION["Brazil Albers",
      METHOD["Albers Equal Area",
        ID["EPSG",9822]],
      PARAMETER["Latitude of false origin",-12,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8821]],
      PARAMETER["Longitude of false origin",-54,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8822]],
      PARAMETER["Latitude of 1st standard parallel",-2,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8823]],
      PARAMETER["Latitude of 2nd standard parallel",-22,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8824]],
      PARAMETER["Easting at false origin",5000000,
        LENGTHUNIT["metre",1],
        ID["EPSG",8826]],
      PARAMETER["Northing at false origin",10000000,
        LENGTHUNIT["metre",1],
        ID["EPSG",8827]]],
    CS[Cartesian,2],
    AXIS["(E)",east,
      ORDER[1],
      LENGTHUNIT["metre",1]],
    AXIS["(N)",north,
      ORDER[2],
      LENGTHUNIT["metre",1]],
    USAGE[
      SCOPE["Statistical analysis."],
      AREA["Brazil - onshore."],
      BBOX[-33.78,-74.01,5.28,-34.74]],
    ID["EPSG",10857]]
```

PROJ exemple

```
+proj=aea +lat_0=-12 +lon_0=-54 +lat_1=-2 +lat_2=-22 +x_0=5000000
+y_0=10000000 +ellps=GRS80 +towgs84=0,0,0,0,0,0,0 +units=m +no_defs +type=crs
```



QUIZ



Why does the Brazil Data Cube use a common projected CRS instead of keeping images in their original spatial reference systems?

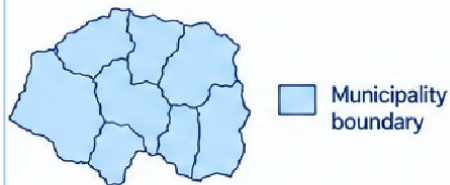
- A. To organize multi-sensor observations within a consistent spatial grid
- B. To increase the spatial resolution of every satellite image
- C. To remove atmospheric effects from the images
- D. To avoid temporal compositing

Spot the problem

You want to estimate land-cover area by municipality.

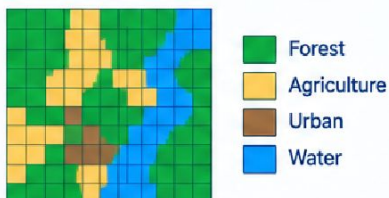


Municipal boundaries



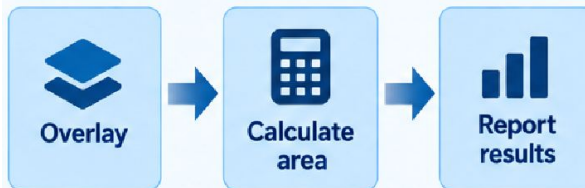
Vector · EPSG:4326 · Geographic CRS

Land-cover raster



Projected CRS · 30 m cells
Brazil Data Cube: EPSG:10857

Analysis workflow



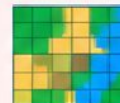
What could go wrong?

1 Overlay the datasets without checking their CRS



EPSG:4326
(geographic)

≠



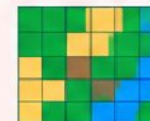
EPSG:10857
(projected)

2 Calculate area directly from longitude/latitude coordinates

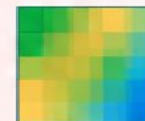


1° of longitude
covers a different
distance at different
latitudes.

3 Resample the categorical raster using bilinear interpolation



Original (categorical)



After bilinear resampling
(blurred classes)

What should you take away?

1

Geographic phenomena must be represented

Data models are abstractions of the geographic world.

2

Raster and vector are analytical choices

The appropriate representation depends on the phenomenon, scale, and question.

3

Coordinates require a reference system

Coordinates alone do not fully define geographic location.

4

Representation affects analysis

Resolution, CRS, projection, and resampling can change measurements and results.



Spatial Representation and CRS

Coordinate systems, datums, projections,
vector, raster

Édipo Cremon
Rennan Marujo

